

CODE APHA ML INTERNSHIP

Stages of the system development

Choosing and obtaining the dataset

For the task 4 (Disease Prediction from medical data) we made use of the Random Forest Algorithm. Why?

Because – the

For the data we will be making use of the Pima Indians Diabetes Dataset (UCI Machine Learning Repository) – this was special designed for learning Supervised Machine Learning.

So, download the CSV file form the GitHub link.

Save the directly inside your main project directory you can rename it to whatever you like (e.g. Ddataset.csv).

Understanding what we are looking at and working with

The dataset contains nine (9) columns which are [Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age, Outcome].

These can be further divided into two main paths

Features (X) – this are the inputs/ attributes we give the model that it uses to learn patterns and makes its decisions (The Known).

And

Target (y) – The ground – truth answers or label we want the model to predict (Unknown) – usually.

Loading and inspecting the data in the main system model (core)

Since the actual Raw SCV did not have column headers built in – we added one by importing pandas and adding a refined explicit list of column names as a list next, we explore the

dataset and check for missing data or invalid entries (in case we have NULL or Missing values in specific columns)

We check of Zeros in each Column if there are any, we fix that by writing a loop that will replace the zeros with the median value of that column (Replacing Missing values with the Median of that column is safer because it prevents extreme outliers from distorting the data.

Data Separation

Before the model is trained, we need to split our dataset into two distinct parts

Features (X) and Target (y).

Now the data is all prepared we split the data into Training and Testing sets.

- The Training Set (Usually 80%)
- The Testing Set (Usually 20%) Next

The actual Machine learning part using a Random Forest Classifier, we get the model setup and trained now we make predictions and grade the model.

After all the evaluation and finetuning we got an accuracy level of a 74.6% as our first evaluation score.

Since we already have a working baseline accuracy of 74.6% the next step is to look under the hood of the random forest algorithm and extract its features importance.

We try to keep track of which features consistently provide the best split in the data.

We use it to determine which feature contributes the most to getting the best possible output.

```
Problems  Output  Debug Console  Terminal

--- Feature Importance ---
      Feature  Importance
1      Glucose  0.328341
5        BMI  0.180409
7        Age  0.160373
4      Insulin  0.101337
6 DiabetesPedigreeFunction  0.073871
0      Pregnancies  0.059946
3      SkinThickness  0.055305
2      BloodPressure  0.040418
```

Now we adjust the algorithm to see if we can squeeze out better performance (Hyperparameter Tuning) by adjusting the `n_estimators` and `max_depth`

Now our accuracy increased to a 75.32.

🚀 Saving and Exporting the model

We use a python library called (joblib) we can now deploy or add a frontend interface.